

Vladimir Baikalov

Research Engineer

Generative Retrieval, Continual Learning, Foundation Recommender Systems

nonameuntitled159@gmail.com

linkedin.com/in/noname-untitled

github.com/NonameUntitled

[Google Scholar Link](#)

Profile

Research engineer building foundation-scale generative recommender and search/retrieval systems that integrate multimodal signals, adapt under non-stationarity, and optimize long-horizon user outcomes.

5+ years of production ML experience across Google/YouTube, Yandex, AI VK, and Huawei, taking models from research prototypes to large-scale deployment. PhD candidate with publications at WWW, RecSys, SIGIR, KDD.

Skills: Python, C++, PyTorch, JAX, TensorFlow, Distributed ML, GPU Acceleration, Large-Scale Data Pipelines, Docker, Kubernetes.

Research Interests: Generative Retrieval, Foundation RecSys, World Models, Bayesian Statistics, Continual Learning, Offline RL, Long-Term Objectives, Multimodal Representation Learning.

Work experience

Lead RecSys Scientist/[AI VK](#)

Dec 2025 – Present

- Leading research on foundation recommender systems for a large video platform, with a focus on generative retrieval, continual semantic updates, and reward-aware long-horizon optimization.
- Co-authored two research works on adaptive generative recommendation: mitigating Semantic ID staleness in generative retrieval and off-policy optimization for long-horizon objectives in large action spaces.
- Developing generative retrieval models for video service and preparing an online A/B test in collaboration with the product team.
- Developing a unified recommender foundation model that combines generative modeling, LLM × RecSys, and reward-aware training.

Lead, R&D RecSys Team/[Yandex Technologies](#)

Jun 2024 – Nov 2025

- Led a three-person R&D team building generative end-to-end recommender models on semantic IDs as an alternative to classic multi-stage retrieval/ranking pipelines.
- Co-developed Argus, increasing Total Listening Time at Yandex.Music by +2.26% It was later adapted to other Yandex products (e.g., e-commerce +1.7% GMV, food-tech services +2.1% GMV).
- Full-stack AI engineer: Owned entire end-to-end ML pipeline from data collection to inference. Built data collection, training, evaluation, and deployment for retrieval & ranking models.

Machine Learning Engineer/[Google, YouTube](#)

Jan 2023 – Jun 2024

- Developed and launched a detection model for **YouTube Shorts**, improving recall by 4% without precision degradation and enabling more frequent fine-tuning of content-quality pipelines.
- Designed an end-to-end serving system for this classifier in collaboration with the **YouTube Gaming** team, taking the model from research prototype to production deployment.
- Worked on production deployment of LLM-based systems, including improvements to fine-tuning and distillation pipelines for large models.
- Built ML systems under **Google engineering practices**, including design/code reviews, production reliability constraints, and cross-team collaboration on large codebases.

Deep Learning Engineer/[Yandex Technologies](#)

Aug 2021 – Dec 2022

- Ported a research PyTorch model into a production C++ ML framework for Ads, increasing GMV by up to 1.5% and CTR by up to 5%.
- Built a distributed parallel data-processing pipeline, improving GPU utilization by up to 2.5 times.

Machine Learning Engineer/[Huawei R&D Dept.](#)

Mar 2020 – Jun 2021

Education

PhD Student/ITMO University

Sep 2024 – Present

Research: Methods for training, scaling, and updating representations in generative recommender systems.

Master Degree/Skoltech, DS major

Sep 2021 – Jul 2024

Honors Thesis: Enhancing recommender systems performance with multiple data representations.

Bachelor Degree/ITMO University, CS major

Sep 2017 – Aug 2021

Thesis: Application of Multi-Agent Routing Algorithms Based on Policy Optimization.

Publications

KDD'26

Scaling Recommender Transformers to One Billion Parameters

Kirill Khrylchenko, Artem Matveev, Sergei Makeev, [Vladimir Baikalov](#)

Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE

Artem Matveev, Sergei Makeev, Aleksei Krasilnikov, [Vladimir Baikalov](#), Sergei Liamaev, Kirill Khrylchenko

SIGIR'26

Mitigating Collaborative Semantic ID Staleness in Generative Retrieval

[Vladimir Baikalov](#), Iskander Bagautdinov, Sergey Muravyov

RecSys'25

Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval

Kirill Khrylchenko, [Vladimir Baikalov](#), Sergei Makeev, Artem Matveev, Sergei Liamaev

Yambda-5B: A Large-Scale Multi-Modal Dataset for Ranking and Retrieval

Alexander Ploshkin, Vladislav Tytskiy, Alexey Pismenny, [Vladimir Baikalov](#), Evgeny Taychinov, Artem Permiakov, Daniil Burlakov, Eugene Krofto

Blending Sequential Embeddings, Graphs, and Engineered Features: 4th Place Solution in RecSys Challenge 2025

Sergei Makeev, Alexandr Andreev, [Vladimir Baikalov](#), Vladislav Tytskiy, Aleksei Krasilnikov, Kirill Khrylchenko

WWW'24

End-to-End Graph-Sequential Representation Learning for Accurate Recommendations

[Vladimir Baikalov](#), Evgeny Frolov

Elsevier'21

Multi-Agent Deep Reinforcement Learning-Based Algorithm for Fast Generalization on Routing Problems

Ibraheem Barbahan, [Vladimir Baikalov](#), Valeriy Vyatkin, Andrey Filchenkov

Teaching Experience

Teaching Assistant/Yandex School of Data Analysis (YSDA)

2024 – 2026

Recommender System Course. [GitHub link \(140+ stars\)](#)

Teaching Assistant/Higher School of Economics

2025 – 2026

Recommender System Course. [GitHub link \(90+ stars\)](#)

Lecturer/ITMO University

2024 – 2026

Taught Programming Massively Parallel Processors.

Topics include: GPU architecture, CUDA, Triton, Kernels, Inference Acceleration, Large-Scale Deep Learning.